

Linear systems

Question 1 / 10 points / Instructor reference

Solve this system using row reduction. Show and label each row operation.

Write the final answer as a column vector.

$$2x + y = 5$$

$$x - y = 1$$

Instructor worked solution

Start

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right]$$

Swap R1 and R2

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & 1 & 5 \end{array} \right]$$

$R2 \leftarrow R2 - 2R1$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 3 \end{array} \right]$$

$R2 \leftarrow (1/3)R2$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array} \right]$$

$R1 \leftarrow R1 + R2$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Therefore, the solution vector is

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Proof by induction

Question 2 / 10 points / Instructor reference

Prove the following statement for every integer $n \geq 1$ by induction.

$$1 + 3 + \dots + (2n - 1) = n^2$$

State the inductive hypothesis for arbitrary $k \geq 1$ and finish with an explicit conclusion that the statement holds for every integer $n \geq 1$.

Instructor worked solution

Base case

At $n = 1$, the sum is $1 = 1^2$, so the statement holds.

Inductive hypothesis

Assume the statement holds for an arbitrary integer $k \geq 1$.

Inductive step

Add the next odd number, $2k + 1$, to the sum for k :

$$k^2 + (2k + 1) = (k + 1)^2.$$

Conclusion

By induction, the statement holds for every integer $n \geq 1$.